

课程笔记

K-means, GMM, EM Algorithm

Codex 基于课程页、公开视频字幕、coverage ledger 与补充官方材料整理

2026 年 4 月 16 日

Why we use the EM algorithm?



- We usually use a **maximum likelihood estimation** to estimate θ
$$\hat{\theta} = \operatorname{argmax}_{\theta} p(O|\theta)$$
- Introduction of **latent variable** (S in this example) makes the problem of estimating θ very difficult

$$p(O|\theta) = \sum_S p(O, S|\theta)$$

- HMM, GMM
- Instead, we use an **expectation-maximization (EM)** algorithm

$$Q(\theta, \theta') = \sum_S p(S|O, \theta') \log p(O, S|\theta)$$

- θ' is obtained by a previous iteration (iterative algorithm)

视频作者/频道: WAVLab

发布日期: 2023-09-20

视频时长: unknown

视频链接: <https://www.youtube.com/watch?v=-1jTd8zM6r4>

目录

1	Hidden Markov Models	2
1.1	K-means, GMM, EM Algorithm	2
1.2	本章小结	3
2	总结与延伸	3

Source Discrepancy

The official course page lists K-means/GMM/EM on 2023-09-20, but the public recording and transcript open with Hidden Markov Models. Treat this as a source-title discrepancy in the record system.

1 Hidden Markov Models

根据课程页主题、公开视频字幕以及本讲的 coverage ledger, Hidden Markov Models 这一部分主要围绕 K-means, GMM, EM Algorithm | K-means | GMM | EM algorithm 展开。写作上保持 coverage-first 原则: 先把老师在这一阶段真正讲过的问题设定、模型部件、训练或推理逻辑讲清, 再压缩成适合复习的结构。

K-means, GMM, EM Algorithm Coverage Map

Coverage-first lecture roadmap generated from coverage_units.jsonl

07-u01 | Hidden Markov Models
K-means, GMM, EM Algorithm | K-means | GMM |
EM algorithm

图 1: 本讲 coverage map: 按 coverage_units.jsonl 归纳出的核心主题与章节映射。

K-means, GMM, EM Algorithm Segment Timeline

Time-aligned topic summary derived from transcript and coverage units

2023-09-20
K-means, GMM, EM Algorithm | K-means | GMM | EM
algorithm

图 2: 本讲 timeline: 根据字幕时间范围归纳的主题推进顺序。

1.1 K-means, GMM, EM Algorithm

从证据位置看, 这一部分对应的课程证据区间是 2023-09-20。课程在这里强调的核心内容可以概括为: K-means, GMM, EM Algorithm | K-means | GMM | EM algorithm。从课程组织上看, 这一部分承担的是把局部概念嵌回整条识别流水线中的作用, 避免读者把公式或模块孤立地记忆。

记录系统中的附加说明指出: The official course page lists K-means/GMM/EM on 2023-09-20, but the public recording title and transcript both open with Hidden Markov Models material. Keep this as a resolved source-title discrepancy in the record system.

在本章结构中，这一单元被并入“Hidden Markov Models”，目的是把同一阶段的定义、机制和权衡集中讲清。

1.2 本章小结

Hidden Markov Models 这一部分把 K-means, GMM, EM Algorithm 中相邻的教学单元串成了一条连续叙事：既交代了这一阶段要解决的问题，也交代了方法、推理或工程权衡为什么要这样组织。

2 总结与延伸

K-means, GMM, EM Algorithm 这一讲在整门《Speech Recognition and Understanding》课程中的位置，是把 K-means, GMM, EM Algorithm 讲成一个可复用的建模模块。从教材化角度看，本讲最重要的不是记住单一术语，而是理解它如何嵌入完整的 automatic speech recognition pipeline：输入语音如何被表示、模型如何被训练、解码如何得到最终转写，以及这些设计分别带来什么假设和权衡。由于公开源以课程页和公开视频字幕为主、缺少官方 slide PDF，本章在正文里尽量把术语、方法和工程语境解释完整，并把缺失项留在 omission 和 manifest 里，而不是凭空补写。